DYNAMIC PHASE TRANSITIONS IN A PARABOLIC PDE SYSTEM

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Abstract. Using dynamic phase transition theory, we explore and classify the phase transitions between stable states in a dissipative parabolic partial differential equation when the control parameter crosses a certain critical threshold. In particular, we first develop a method to compute a leading-order approximation of a center manifold to which we can reduce any PDE of interest. This reduced equation can then be analyzed at the critical value of the control parameter to classify a phase transition in one of three categories: continuous, jump, or mixed. Using the blow-up method to investigate degenerate fixed points, we find that for most tuplets of coefficients, the parabolic system undergoes a jump or mixed transition.

1. Introduction to dynamic transition theory

We begin with an overview of general dynamic transition theory. Let X and X_1 be two Banach spaces, with $X_1 \subset X$ a compact and dense inclusion. We will consider nonlinear evolution equations in the form

$$\frac{du}{dt} = L_\lambda u + G(u, \lambda), \qquad u(0) = u_0 \tag{1}$$

Here, $\lambda \in \mathbb{R}$ represents a control parameter. We also assume that $L_\lambda : X_1 \to X$ is a parametrized linear completely continuous field depending on λ, satsifying

$$\begin{aligned} &L_\lambda = -A + B_\lambda && \text{is a sectorial operator} \\ &A : X_1 \to X && \text{is a linear homeomorphism, and} \\ &B_\lambda : X_1 \to X && \text{is a linear compact operator} \end{aligned} \tag{2}$$

We can then define fractional power operators L_λ^α for $\alpha \in \mathbb{R}$ with domains $X_\alpha = D(L_\lambda^\alpha)$. Assume that $G(\cdot, \lambda) : X_\alpha \to X$ is a C^r $(r \geq 1)$ bounded mapping for some $0 \leq \alpha < 1$, which depends continuously on λ. Additionally, assume that

$$G(u, \lambda) = o(\|u\|_{X_\alpha}) \quad \forall \lambda \in \mathbb{R} \tag{3}$$

We will always assume the conditions 2 and 3, which imply that the system (1) has a dissipative structure.

Definition 1. [MW19] The system (1) is said to undergo a dynamic transition from $(u, \lambda) = (0, \lambda_0)$ at $\lambda = \lambda_0$ if the following conditions hold:

(1) if $\lambda < \lambda_0$, $u = 0$ is locally asymptotically stable for (1)
(2) if $\lambda > \lambda_0$, there exists a neighborhood $U \subset X$ of $u = 0$ independent of λ, such that for any $u_0 \in U \setminus \Gamma_\lambda$, the solution $u_\lambda(t, u_0)$ of (1) satisfies that

$$\limsup_{t \to \infty} \|u_\lambda(t, u_0)\| \geq \delta(\lambda) > 0, \qquad \lim_{\lambda \to \lambda_0} \delta(\lambda) \geq 0,$$

where Γ_λ is the stable manifold of $u = 0$ with $\operatorname{codim}(\Gamma_\lambda) \geq 1$ in X for $\lambda > \lambda_0$.

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We will now outline the principle of exchange of stabilities (PES), which describes a system in which some number of critical eigenvalues transition from having negative real part to positive real part at the critical value $\lambda = \lambda_0$. Denote the generalized eigenvalues, counting multiplicity, of L_λ by $\{\beta_j(\lambda) \in \mathbb{C} \mid j = 1, 2, \ldots\}$. Such a situation must satisfy the following conditions:

$$\begin{aligned} &\mathrm{Re}\beta_i(\lambda) \begin{cases} < 0 & \text{if } \lambda < \lambda_0, \\ = 0 & \text{if } \lambda = \lambda_0, \\ > 0 & \text{if } \lambda > \lambda_0 \end{cases} && \forall 1 \leq i \leq m \\ &\mathrm{Re}\beta_j(\lambda_0) < 0 && \forall j \geq m+1 \end{aligned} \tag{4}$$

These conditions form the basis for the PES.

Theorem 1. [MW19] *Assume that the conditions in (4) hold true. Then, the system (1) always undergoes a transition from $(u, \lambda) = (0, \lambda_0)$ at $\lambda = \lambda_0$, and there is a neighborhood $U \subset X$ of $u = 0$ such that the transition is one of the following three types.*

(1) Continuous Transition: *there exists an open and dense set $\tilde{U}_\lambda \subset U$ such that for any $u_0 \in \tilde{U}_\lambda$, the solution $u_\lambda(t, u_0)$ of (1) satisfies*

$$\lim_{\lambda \to \lambda_0} \limsup_{t \to \infty} \|u_\lambda(t, u_0)\| = 0.$$

(2) Jump Transition: *for any $\lambda_0 < \lambda < \lambda_0 + \varepsilon$ for some $\varepsilon > 0$, there is an open and dense set $U_\lambda \subset U$ such that for any $u_0 \in U_\lambda$,*

$$\limsup_{t \to \infty} \|u_\lambda(t, u_0)\| \geq \delta > 0,$$

where $\delta > 0$ is independent of λ.

(3) Mixed Transition: *for any $\lambda_0 < \lambda < \lambda_0 + \varepsilon$ for some $\varepsilon > 0$, U can be decomposed into two open, but not necessarily connected, sets U_1^λ and U_2^λ, where $\overline{U} = \overline{U}_1^\lambda \cup \overline{U}_2^\lambda$, $U_1^\lambda \cap U_2^\lambda = \varnothing$, such that*

$$\begin{aligned} &\lim_{\lambda \to \lambda_0} \limsup_{t \to \infty} \|u(t, u_0)\| = 0 && \forall u_0 \in U_1^\lambda, \\ &\limsup_{t \to \infty} \|u(t, u_0)\| \geq \delta > 0 && \forall u_0 \in U_2^\lambda, \end{aligned}$$

where U_1^λ and U_2^λ are called metastable domains.

In a dissipative system, there always exists some stable state, represented by a local attractor. A local attractor is defined as follows:

Definition 2. A set $A \subset X$ is considered a local attractor for the system (1) at λ if there exists an open set $U \subset X$ such that

$$A = \bigcap_{t \geq 0} \Bigg(\bigcup_{\substack{s \geq t \\ u_0 \in U}} u(s, u_0) \Bigg).$$

In a dynamic transition satisfying the conditions in 4, we call the stable states at values of $\lambda > \lambda_0$ transition states. We can visually interpret Theorem 1 in terms of these transition states, which are local attractors. In a continuous transition, as λ crosses λ_0, the equilibrium point at $u = 0$ deforms continuously into a local attractor near the origin. In a jump transition, as λ crosses λ_0, the transition state "jumps" discontinuously to a local attractor

far from the origin. In a mixed transition, the equilibrium point deforms continuously into a local attractor, but that local attractor does not attract an entire dense open set about the origin.

In practice, it is somewhat difficult to analyze the behavior of solutions for the system (1) for a range of control parameters $\lambda \in (\lambda_0 - \varepsilon, \lambda_0 + \varepsilon)$. The following theorem allows us to make some conclusions about the transition type of the system based only on its behavior at $\lambda = \lambda_0$.

Theorem 2. [MW19] *Let the conditions in (4) hold true. Then, the transition type of (1) is completely determined by its reduced equation to the center manifold (see Section 2). In particular, let $U \subset \mathbb{R}^m$ be a neighborhood of $z = 0$. Then we have the following assertions:*

(1) *If the transition of (1) at $(0, \lambda_0)$ is continuous, then there is an open and dense set $\tilde{U} \subset U$ such that for any $z_0 \in \tilde{U}$ the solution $z(t, z_0)$ of the reduced equation at $\lambda = \lambda_0$ satisfies that*
$$\lim_{t\to\infty} z(t, z_0) = 0.$$
(2) *If there exists an open and dense set $\tilde{U} \subset U$ such that for any $z_0 \in \tilde{U}$ the solution $z(t, z_0)$ of the reduced equation satisfies*
$$\limsup_{t\to\infty} |z(t, z_0)| \neq 0,$$
then the transition is a jump transition.
(3) *If the transition is mixed, then there exist two open sets $\tilde{U}_1, \tilde{U}_2 \subset U$ such that*
$$\lim_{t\to\infty} z(t, z_0) = 0 \qquad \forall z_0 \in \tilde{U}_1$$
$$\limsup_{t\to\infty} |z(t, z_0)| \neq 0 \qquad \forall z_0 \in \tilde{U}_2.$$

2. Center manifold approximation

In this section, we analyze the system (1) from Section 1, which we will restate here:
$$u_t = L_\lambda u + G(u, \lambda)$$
where $L_\lambda : X_1 \to X$ is a linear operator and G is a nonlinear operator. The conditions on L_λ and G are stated more precisely in Section 1. Let G have the Taylor expansion at $u = 0$ as follows:
$$G(u, \lambda) = \sum_{\ell=n}^{r} G_\ell(u, \lambda) + o(\|u\|_{X_1}^r) \qquad \text{for some } 2 \leq n \leq r \tag{5}$$
Here, each G_ℓ is an ℓ-multiple linear operator, with G_n being the leading order term of this Taylor series. Let us assume conditions in (4) for a transition at $\lambda = \lambda_0$. Thus, we have the critical eigenvectors $\{e_1(\lambda), \ldots, e_m(\lambda)\}$ and $\{e_1^*(\lambda), \ldots, e_m^*(\lambda)\}$ of L_λ and the conjugate operator L_λ^*, respectively, corresponding to the eigenvalues $\beta_i(\lambda)$ $(1 \leq i \leq m)$. Now, by the spectral theorem, stated in Theorem 3.4 of [MW05], the spaces X_1 and X can be decomposed into a direct sum:
$$X_1 = E_1^\lambda \bigoplus E_2^\lambda, \qquad X = E_1^\lambda \bigoplus \overline{E}_2^\lambda,$$

where

$$E_1^\lambda = \text{span}\{e_1(\lambda), \dots, e_m(\lambda)\},$$
$$E_2^\lambda = \{u \in X_1 \mid (u, e_i^*(\lambda)) = 0 \quad \forall 1 \le i \le m\}$$
$$\overline{E}_2^\lambda = \text{closure of } E_2^\lambda \text{ in } X$$

This allows us to naturally define a decomposition of the linear operator L_λ:

$$L_\lambda = J_\lambda \bigoplus \mathcal{L}_\lambda \tag{6}$$

Here, $J_\lambda = L_\lambda\big|_{E_1^\lambda} : E_1^\lambda \to E_1^\lambda$, which, under a generalized eigenbasis, is a Jordan matrix. $\mathcal{L}_\lambda = L_\lambda\big|_{E_2^\lambda} : E_2^\lambda \to \overline{E}_2^\lambda$.
From [Hen81], we know that a center manifold for the system (2) exists and, near $x = 0$, satisfies

$$\Phi(x, \lambda) = \int_{-\infty}^{0} e^{-\tau \mathcal{L}_\lambda} P_2 G(z(\tau, x) + \Phi(z(\tau, x)), \lambda)\, d\tau \tag{7}$$

Then, we can reduce the system (2) to the following form near the origin:

$$\frac{dz}{dt} = J_\lambda z + P_1 G(z + \Phi(z), \lambda), \;\; z(0) = x \tag{8}$$

Assume that J_λ is a simple Jordan block near $\lambda = \lambda_0$

$$J_\lambda = \begin{pmatrix} \beta(\lambda) & \delta & \cdots & 0 & 0 \\ 0 & \beta(\lambda) & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & \beta(\lambda) & \delta \\ 0 & 0 & \cdots & 0 & \beta(\lambda) \end{pmatrix} \tag{9}$$

Now, we can state the main theorem for this section. This theorem is an extension of Theorem A.1.1 in [MW19].

Theorem 3. *Assume that the conditions in (4) hold true, and $G(u, \lambda)$ has the Taylor expansion as in (5). Then, the center manifold function $\Phi : E_1 \to E_2$ of (2) near $\lambda = \lambda_0$ can be expressed as*

$$\Phi(x, \lambda) = \int_{-\infty}^{0} e^{-\tau \mathcal{L}_\lambda} P_2 G_n(e^{\tau J_\lambda} x, \lambda)\, d\tau + o(\|x\|^n) \tag{10}$$

Additionally, if J_λ is in the form given by (9), the center manifold function is given by

$$\Phi(x, \lambda) = \sum_{k=1}^{2m-1} \phi_k(x, \lambda) + o(\|x\|^n) + \mathcal{O}(|Re(\beta)|\|x\|^n), \quad -\mathcal{L}_\lambda^k \phi_k = (k-1)! P_2 F_k(x, \lambda) \tag{11}$$

where

$$F_k(x, \lambda) = \delta^{k-1} \sum_{\substack{\sum_{j=1}^n i_j = k-1 \\ 0 \le i_1, \dots, i_n < m}} \frac{1}{\prod_{j=1}^n i_j!} G_n(M_{i_1} x, \dots, M_{i_j} x, \lambda) \tag{12}$$

Proof. We will show the proof in full for the example where $n = 2$, meaning that $G(u, \lambda)$ has a leading-order bilinear term $G_2(u, \lambda)$.

From Appendix A of [MW19], the reduced equation (8) has a solution given by

$$z(t) = \sum_{j=1}^{m} z_j e_j + o(\|x\|), \quad z_j = \xi_j(t) e^{\beta(\lambda)t}, \quad \xi_j(t) = \sum_{r=0}^{m-j} \frac{\delta^r t^r x_{j+r}}{r!} \tag{13}$$

Near $\lambda = \lambda_0$, $\beta(\lambda)$ is nearly zero, so we can approximate

$$z(t) = \sum_{j=1}^{m} \xi_j e_j + o(1) = \begin{pmatrix} x_1 \\ \vdots \\ x_{m-2} \\ x_{m-1} \\ x_m \end{pmatrix} + \delta t \begin{pmatrix} x_2 \\ \vdots \\ x_{m-1} \\ x_m \\ 0 \end{pmatrix} + \frac{\delta^2}{2} t^2 \begin{pmatrix} x_3 \\ \vdots \\ x_m \\ 0 \\ 0 \end{pmatrix} + \cdots + o(1) \tag{14}$$

Here, $o(k) = o(\|x\|^k) + O(|\mathrm{Re}(\beta)|\|x\|^k)$. We can rewrite the previous expression (14) as

$$z(t) = \sum_{k=0}^{m-1} \delta^k t^k M_k x + o(1) \tag{15}$$

Here, M_k represents an $m \times m$ matrix with

$$(M_k)_{ij} = \begin{cases} 1, & j - i = k \\ 0, & j - i \neq k \end{cases}$$

Therefore,

$$\begin{aligned} G_2(z, z, \lambda) &= \sum_{0 \le i,j < m} G_2(\delta^i t^i M_i(x), \delta^j t^j M_j(x)) + o(2) \\ &= \sum_{k=1}^{2m-1} t^{k-1} F_k(x), \lambda + o(2) \end{aligned}$$

Where

$$F_k(x, \lambda) = \delta^{k-1} \sum_{0 \le j < k} \frac{1}{j!(k-1-j)!} G_2(M_j x, M_{k-1-j} x, \lambda) \tag{16}$$

Then, as in Appendix A of [MW19], we can derive that

$$\Phi(x, \lambda) = \sum_{k=1}^{2m-1} \phi_k + o(n), \quad -\mathcal{L}_\lambda^k \phi_k = (k-1)! P_2 F_k(x, \lambda) \tag{17}$$

In the case that $\mathcal{L}_\lambda$ is symmetric, we can go even further, saying that

$$\begin{aligned} \phi_k(x, \lambda) &= \sum_{j > m} \phi_{kj}(x, \lambda) e_j \\ \phi_{kj}(x, \lambda) &= -\frac{(k-1)!}{\beta_j(\lambda)^k (e_j, e_j^*)} (F_k(x, \lambda), e_j^*) \end{aligned} \tag{18}$$

So in the case that $\mathcal{L}_\lambda$ is symmetric, we can express $\Phi(x, \lambda)$ as a double summation.

$$\Phi(x, \lambda) = \sum_{k=1}^{2m-1} \sum_{j > m} \phi_{kj}(x, \lambda) e_j + o(n) \tag{19}$$

In the case in which $n > 2$, we can use the same steps as above, with some slight modifications. Using the same definitions of x, z, and M_k, we can find that

$$G_n(z,\lambda) = \sum_{0\le i_1,\dots,i_n<m} G_n(M_{i_1}x,\dots,M_{i_n}x,\lambda) = \sum_{k=1}^{n(m-1)+1} t^{k-1}F_k(x,\lambda),$$

where we can define $F_k(x,\lambda)$ as follows:

$$F_k(x,\lambda) = \delta^{k-1} \sum_{\substack{\sum_{j=1}^n i_j = k-1 \\ 0\le i_1,\dots,i_n<m}} \frac{1}{\prod_{j=1}^n i_j!} G_n(M_{i_1}x,\dots,M_{i_j}x,\lambda) \tag{20}$$

Like before, we can calculate the center manifold functions using these $F_k(x)$ using Equation 17. In practice, this procedure becomes very computationally expensive as n increases, involving m^n evaluations of G_n to get the level of precision given in Equation 2. □

Note that in this proof, we could find a double summation for $\Phi(x,\lambda)$ if $\mathcal{L}_\lambda$ is symmetric. We can do a similar process for $\mathcal{L}_\lambda$ in Jordan form and decompose $\Phi(x,\lambda)$ into a double sum in the form of (19); however, in this case, it is more difficult to get a simple explicit formula for each ϕ_{kj}. However, one example of this procedure is shown in Section 3.

Given Equation 19 for the center manifold, we can explicitly find the reduced equation up to a certain number of terms. In particular, if $G(u,\lambda) = B(u,\lambda)$ is a bilinear form,

$$\begin{aligned} \frac{dz_\ell}{dt} &= J_\lambda(z_\ell e_\ell) + \sum_{1\le i,j\le m} z_i z_j \frac{\big(B(e_i,e_j,\lambda), e_\ell^*\big)}{(e_\ell, e_\ell^*)} \\ &\quad + \sum_{\substack{1\le i\le m \\ j>m}} z_i \left(\sum_{k=1}^{2m-1} \phi_{kj}(z,\lambda) \right) \frac{\big(B(e_i,e_j,\lambda) + B(e_j,e_i,\lambda), e_\ell^*\big)}{(e_\ell, e_\ell^*)} + o(\|z\|^3) \end{aligned} \tag{21}$$

for $1 \le \ell \le m$.

3. Application to a parabolic system

This section addresses the following parabolic system:

$$\begin{aligned} \frac{\partial u_1}{\partial t} &= (u_1)_{xx} + \lambda u_1 + a u_2 + a_{11}u_1^2 + a_{12}u_1u_2 + a_{22}u_2^2 - K_1 u_1^3 \\ \frac{\partial u_2}{\partial t} &= (u_2)_{xx} + \lambda u_2 + b_{11}u_1^2 + b_{12}u_1u_2 + b_{22}u_2^2 - K_2 u_2^3 \\ \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}\Bigg|_{x\in\{0,\pi\}} &= 0 \end{aligned} \tag{22}$$

This system has applications in chemistry and biology, particularly in reaction-diffusion systems. Reaction-diffusion systems are characterized by constituents that undergo chemical reactions and disperse in space by diffusion. This system is also of particular interest in dynamic phase transition theory, because, as we will soon discover, each eigenvalue of the system has algebraic multiplicity 2, but geometric multiplicity 1. This is in contrast to the simpler scenario where the operator L_λ is symmetric, having only ordinary eigenvectors.

We begin by finding the (generalized) eigenfunctions and eigenvalues for the linear part of this PDE system. Let L_λ be the linear operator defined by

$$L_\lambda \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} (u_1)_{xx} + \lambda u_1 + a u_2 \\ (u_2)_{xx} + \lambda u_2 \end{pmatrix} \tag{23}$$

We can see that the (generalized) eigenfunctions and eigenvalues of L_λ are given by

$$\begin{aligned} e_{j1} = e_{j1}^* &= \begin{pmatrix} \sin(jx) \\ 0 \end{pmatrix} & e_{j2} = e_{j2}^* &= \begin{pmatrix} 0 \\ \sin(jx) \end{pmatrix} \\ \beta_{j1}(\lambda) &= \lambda - j^2 & \beta_{j2}(\lambda) &= \lambda - j^2 \end{aligned} \tag{24}$$

for $j \in \mathbb{Z}_+$. Note that the e_{j1} are ordinary eigenvectors, but the e_{j2} are generalized eigenvectors of rank 2. It is clear that the critical eigenfunctions are e_{11} and e_{12}, satisfying the conditions of (4) for a dynamical transition at $\lambda = 1$. Thus, J_λ is determined by the action of L_λ on $(e_{11}, e_{12})^t$:

$$J_\lambda = \begin{pmatrix} \lambda - 1 & a \\ 0 & \lambda - 1 \end{pmatrix} \tag{25}$$

In a similar manner, we can determine the action of L_λ on the space spanned by $\{e_{j1}, e_{j2}\}$ for any $j > 1$. Denote this matrix $L_{\lambda,j}$, which takes the form

$$L_{\lambda,j} = \begin{pmatrix} \lambda - j^2 & a \\ 0 & \lambda - j^2 \end{pmatrix} \tag{26}$$

To compute the center manifold function, we need to calculate the $F_k(y, \lambda)$ as defined in Equation 12, where $y = y_1 e_{11} + y_2 e_{12}$. Note that the nonlinear part of the parabolic system (22) is independent of λ. Some simple calculations yield

$$\begin{aligned} F_1(y) &= \begin{pmatrix} a_{11}y_1^2 + a_{12}y_1y_2 + a_{22}y_2^2 \\ b_{11}y_1^2 + b_{12}y_1y_2 + b_{22}y_2^2 \end{pmatrix} \sin^2(x) = \begin{pmatrix} A_1 \\ B_1 \end{pmatrix} \sin^2(x) \\ F_2(y) &= a \begin{pmatrix} 2a_{11}y_1y_2 + a_{12}y_2^2 \\ 2b_{11}y_1y_2 + b_{12}y_2^2 \end{pmatrix} \sin^2(x) = \begin{pmatrix} A_2 \\ B_2 \end{pmatrix} \sin^2(x) \\ F_3(y) &= a^2 \begin{pmatrix} a_{11}y_2^2 \\ b_{11}y_2^2 \end{pmatrix} \sin^2(x) = \begin{pmatrix} A_3 \\ B_3 \end{pmatrix} \sin^2(x) \end{aligned} \tag{27}$$

We can express the center manifold in the following form:

$$\Phi(y, \lambda) = \sum_{k=1}^{3} \phi_k = \sum_{k=1}^{3} \sum_{j>1} \sum_{i=1}^{2} \phi_{kji} e_{ji} \tag{28}$$

To calculate the ϕ_{kji}, we use the fact that $-\mathcal{L}_\lambda^k \phi_k = (k-1)! P_2 F_k(y)$ from Equation 17.

$$-(\mathcal{L}_\lambda^k \phi_k, e_{ji}^*) = -(L_{\lambda,j}^k(\phi_{kj1}e_{j1} + \phi_{kj2}e_{j2}), e_{ji}^*) = (k-1)!(F_k(y), e_{ji}^*)$$

$$\Downarrow$$

$$-((\lambda - j^2)^k \phi_{kj1} e_{j1} + ka(\lambda - j^2)^{k-1} \phi_{kj2} e_{j1} + (\lambda - j^2)^k \phi_{kj2} e_{j2}, e_{ji}^*) = (k-1)!(F_k(y), e_{ji}^*)$$

Splitting into two cases for $i = 1$ and $i = 2$, we find that

$$\begin{aligned} \frac{\pi}{2}((\lambda - j^2)^k \phi_{kj1} + ka(\lambda - j^2)^{k-1} \phi_{kj2}) &= -(k-1)!(F_k(y), e_{j1}^*) \\ \frac{\pi}{2}(\lambda - j^2)^k \phi_{kj2} &= -(k-1)!(F_k(y), e_{j2}^*) \end{aligned}$$

For the inner product in the equations above, we will need to evaluate the following integral; the $\sin^2(x)$ comes from the $F_k(y)$, and the $\sin(jx)$ is from the e^*_{ji}.

$$\int_0^\pi \sin^2(x)\sin(jx) = \begin{cases} -\frac{4}{(j-2)j(j+2)}\,,\ j \text{ odd} \\ 0\,,\ j \text{ even} \end{cases}$$

Using this integral, we can then find an explicit expression for all of the ϕ_{kji}. Note that these equations only hold for j odd; if j is even, $\phi_{kji} = 0$.

$$\begin{aligned} \phi_{kj1} &= \frac{8(k-1)!}{\pi(\lambda - j^2)^k (j-2)j(j+2)} A_k - \frac{8ak!}{\pi(\lambda - j^2)^{k+1}(j-2)j(j+2)} B_k \\ \phi_{kj2} &= \frac{8(k-1)!}{\pi(\lambda - j^2)^k (j-2)j(j+2)} B_k \end{aligned} \tag{29}$$

Note that the A_k and B_k in the expressions above are the same ones defined in Equation 27. This gives us an explicit form for the leading-order approximation of our center manifold function:

$$\Phi(y,\lambda) = \sum_{k=1}^{3} \sum_{\substack{j\geq 3 \\ j \text{ odd}}} (\phi_{kj1} e_{j1} + \phi_{kj2} e_{j2}) + o(2). \tag{30}$$

We can now find the equation for the reduced system at $\lambda = \lambda_0$, using Equation 8. Note that we can also find the reduced equation at values of λ near λ_0, the results of which are described in the Appendix.

$$
\begin{aligned}
y_1' &= ay_2 + \frac{8}{3\pi}(a_{11}y_1^2 + a_{12}y_1y_2 + a_{22}y_2^2) + \Big(C_0(2a_{11}^2 + a_{12}b_{11}) + 2C_1aa_{11}b_{11} - \frac{3}{4}K_1\Big)y_1^3 \\
&+ \Big(C_0(3a_{11}a_{12} + a_{12}b_{12} + 2a_{22}b_{11}) + C_1a(-4a_{11}^2 + 2a_{11}b_{12} - a_{12}b_{11}) - 4C_2a^2a_{11}b_{11}\Big)y_1^2y_2 \\
&+ \Big(C_0(2a_{11}a_{22} + a_{12}^2 + a_{12}b_{22} + 2a_{22}b_{12}) + C_1a(-4a_{11}a_{12} + 2a_{11}b_{22} - 4a_{22}b_{11}) \\
&+ C_2a^2(2a_{11}^2 - 2a_{11}b_{12} - a_{12}b_{11}) + 2C_3a^3a_{11}b_{11}\Big)y_1y_2^2 + \Big(C_0(a_{12}a_{22} + 2a_{22}b_{22}) \\
&+ C_1a(-a_{12}^2 + a_{12}b_{22} - 2a_{22}b_{12}) + C_2a^2(a_{11}a_{12} - a_{12}b_{12} + 2a_{22}b_{11}) + C_3a^3a_{12}b_{11}\Big)y_2^3 \\
&+ o(\|y\|^3) \\
y_2' &= \frac{8}{3\pi}(b_{11}y_1^2 + b_{12}y_1y_2 + b_{22}y_2^2) + \Big(C_0(2a_{11}b_{11} + b_{11}b_{12}) + 2C_1ab_{11}^2\Big)y_1^3 \\
&+ \Big(C_0(a_{11}b_{12} + 2a_{12}b_{11} + 2b_{11}b_{22} + b_{12}^2) + C_1a(-4a_{11}b_{11} + b_{11}b_{12}) - 4C_2a^2b_{11}^2\Big)y_1^2y_2 \\
&+ \Big(C_0(a_{12}b_{12} + 2a_{22}b_{11} + 3b_{12}b_{22}) + C_1a(-2a_{11}b_{12} - 2a_{12}b_{11} - 2b_{11}b_{22}) \\
&+ C_2a^2(2a_{11}b_{11} - 3b_{11}b_{12}) + 2C_3a^3b_{11}^2\Big)y_1y_2^2 + \Big(C_0(a_{22}b_{12} + 2b_{22}^2) + C_1a(-a_{12}b_{12} - b_{12}b_{22}) \\
&+ C_2a^2(a_{11}b_{12} + 2b_{11}b_{22} - b_{12}^2) + C_3a^3b_{11}b_{12} - \frac{3}{4}K_2\Big)y_2^3 + o(\|y\|^3)
\end{aligned}
$$

Here, the constants C_i are defined as

$$
\begin{aligned}
C_0 &= \frac{1}{108}\Big(\frac{448}{\pi^2} - 45\Big) \approx 0.003629 \\
C_1 &= \frac{1}{108}\Big(-\frac{720}{\pi^2} + 73\Big) \approx 0.000451 \\
C_2 &= \frac{1}{486}\Big(\frac{6248}{\pi^2} - 633\Big) \approx 0.000113 \\
C_3 &= \frac{1}{2430}\Big(-\frac{106135}{\pi^2} + 10665 + 9\pi^2\Big) \approx 0.000042
\end{aligned}
$$

4. Dynamic transition classification for the parabolic system

Before attempting to classify the transitions of the parabolic system, let us introduce the blow-up method used to analyze the stability of a 2-dimensional ODE. We begin with the normal form theorem.

Theorem 4. [Tak74, GH83] *Let $x' = f(x)$ be an n-dimensional C^r system of differential equations with $f(0) = 0$ and $Df(0)x = L$. Let H_k denote the vector space of vector fields on $\mathbb{R}^n$ whose coefficient functions are homogeneous polynomials of degree k. Define a splitting $H_k = B_k + G_k$, where $B_k = Im([L, -])$ and G_k is some supplementary space. Then, there*

is an analytic change of coordinates in a neighborhood of the origin which transforms the system $x' = f(x)$ *to* $y' = g(y) = L + g_2(y) + g_3(y) + \cdots + g_r(y) + R_r$*, with* $g_k \in G_k$ *for* $2 \le k \le r$ *and* $R_r = o(|y|^r)$*.*

The proof of Theorem 4 in [Tak74] proceeds by induction: if we have a system in the form $f'(x) = L + g_2(y) + \cdots + g_{\ell-1}(y) + R_{\ell-1}$, where $R_{\ell-1} = g_\ell + b_\ell + R_\ell$, then there is an analytic change of coordinates to a new system $y' = L + g_2(y) + \cdots + g_\ell + R'_\ell$. We can use this normal form approach to simplify the form of the 2-dimensional ODEs we are studying, with $L = \begin{pmatrix} ay_2 \\ 0 \end{pmatrix}$, $a \neq 0$.

In this scenario, we can find B_2 as follows:

(31)
$$\begin{aligned} B_2 &= \operatorname{span}\left\{ \left[L, \begin{pmatrix} y_1^2 \\ 0 \end{pmatrix}\right], \left[L, \begin{pmatrix} y_1y_2 \\ 0 \end{pmatrix}\right], \left[L, \begin{pmatrix} y_2^2 \\ 0 \end{pmatrix}\right], \left[L, \begin{pmatrix} 0 \\ y_1^2 \end{pmatrix}\right], \left[L, \begin{pmatrix} 0 \\ y_1y_2 \end{pmatrix}\right], \left[L, \begin{pmatrix} 0 \\ y_2^2 \end{pmatrix}\right] \right\} \\ &= \operatorname{span}\left\{ \begin{pmatrix} 2y_1y_2 \\ 0 \end{pmatrix}, \begin{pmatrix} y_2^2 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} y_1^2 \\ -2y_1y_2 \end{pmatrix}, \begin{pmatrix} y_1y_2 \\ -y_2^2 \end{pmatrix}, \begin{pmatrix} y_2^2 \\ 0 \end{pmatrix} \right\} \end{aligned}$$

Let us define the four linearly independent vectors as

$$B_2^1 = \begin{pmatrix} 2y_1y_2 \\ 0 \end{pmatrix}, \quad B_2^2 = \begin{pmatrix} y_2^2 \\ 0 \end{pmatrix}, \quad B_2^3 = \begin{pmatrix} y_1^2 \\ -2y_1y_2 \end{pmatrix}, \quad B_2^4 = \begin{pmatrix} y_1y_2 \\ -y_2^2 \end{pmatrix},$$

Thus, a natural definition for G_2 is

$$G_2 = \operatorname{span}\left\{ \begin{pmatrix} y_1^2 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ y_1^2 \end{pmatrix} \right\}, \tag{32}$$

where we can denote

$$G_2^1 = \begin{pmatrix} y_1^2 \\ 0 \end{pmatrix}, \quad G_2^2 = \begin{pmatrix} 0 \\ y_1^2 \end{pmatrix}$$

Thus, for a system in the form

$$\begin{aligned} y_1' &= ay_2 + \tilde{a}_{11}y_1^2 + \tilde{a}_{12}y_1y_2 + \tilde{a}_{22}y_2^2 + O(y^3) \\ y_2' &= \tilde{b}_{11}y_1^2 + \tilde{b}_{12}y_1y_2 + \tilde{b}_{22}y_2^2 + O(y^3), \end{aligned} \tag{33}$$

we can rewrite the second-order terms to get

$$\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} ay_2 \\ 0 \end{pmatrix} + (\tilde{a}_{12} + \tilde{b}_{22})B_2^1 + \tilde{a}_{22}B_2^2 - \frac{1}{2}\tilde{b}_{12}B_2^3 - \tilde{b}_{22}B_2^4 + \left(\tilde{a}_{11} + \frac{\tilde{b}_{12}}{2}\right)G_2^1 + \tilde{b}_{11}G_2^2,$$

so there is an analytic change of coordinates to a new system

$$\begin{aligned} y_1' &= ay_2 + \left(\tilde{a}_{11} + \frac{\tilde{b}_{12}}{2}\right)y_1^2 + O(y^3) \\ y_2' &= \tilde{b}_{11}y_1^2 + O(y^3) \end{aligned} \tag{34}$$

Along with the normal form theorem, we will use the blow-up method in our stability analysis. A full example of this method in practice will be given in the proof of Theorem 5. In essence, the blow-up method is done by converting a system to polar coordinates, and "blowing up" the origin. This is done by viewing (r, θ) as coordinates on a cylinder, where r can take on both positive and negative values, which expands the singularity at the origin into a circle. Such a perspective is illustrated in Figure 1. Then, one can analyze the fixed points of the new system on the circle at $r = 0$ and perform the same blow-up method on

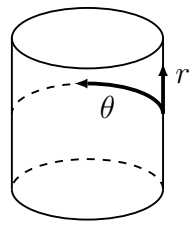

FIGURE 1. Polar coordinates (r, θ) as viewed on a cylinder.

those points. This process continues until, hopefully, all of the fixed points are hyperbolic. If such a stage is reached, one can "blow down" and reverse the process to show that the flows of the differential equation near the origin are independent of some higher-order terms.

We can now analyze the dynamic transition type of this parabolic system in a few special cases. We can compile our results into the following theorem on transition classification for a parabolic PDE system (22).

Theorem 5. *For the parabolic PDE system (22), we can conclude that*

(1) *If $b_{11} \neq 0$, the system undergoes a jump transition at $\lambda = \lambda_0$.*
(2) *If $b_{11} = 0$, $a_{11} \neq 0$, $b_{12} \neq 0$, and $2a_{11} - b_{12} \neq 0$, we can split our situation into two sub-cases. If $a_{11}b_{12} < 0$, then the system undergoes a jump transition. If $a_{11}b_{12} > 0$, then the system undergoes a jump or mixed transition.*

Proof. The proof of this theorem will be done by analyzing two cases, corresponding to the two assertions in the theorem. For convenience, we will denote the quadratic coefficients $\frac{8}{3\pi}a_{11}, \frac{8}{3\pi}a_{12}, \ldots$ as $\tilde{a}_{11}, \tilde{a}_{12}, \ldots$.

CASE 1: $b_{11} \neq 0$.

By Theorem 4, there is a suitable change of coordinates that reduces the general differential equation to one of the following form at $\lambda = \lambda_0$:

$$y_1' = ay_2 + \left(\tilde{a}_{11} + \frac{\tilde{b}_{12}}{2}\right)y_1^2 + O(y^3)$$
$$y_2' = \tilde{b}_{11}y_1^2 + O(y^3)$$

Now, as long as $\tilde{b}_{11} \neq 0$, the blow-up method shows that for fixed a, any system of differential equations in this form has the same local flow in a small neighborhood of the origin. A detailed analysis of this case is presented in both [Tak74] and [GH83]. Thus, for this case, it is sufficient to analyze the behavior of one particular set of coefficients. We will analyze the following, simpler, system:

$$y_1' = ay_2$$
$$y_2' = y_1^2$$

We can re-express this system as a one-dimensional ODE:

$$y_1'' = ay_1^2$$

We can then multiply both sides by y_1' and integrate to find invariant level curves of the system. Finally, substituting $y_1' = ay_2$ gives us level curves of the system, which take the form

$$3ay_2^2 - 2y_1^3 = c$$

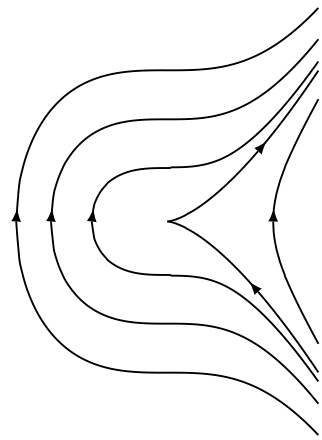

FIGURE 2. Local flows of the reduced equation at $\lambda = \lambda_0$ when $b_{11} \neq 0$.

We can use these invariant trajectories and the fact that $y_2' \geq 0$ to construct a diagram of the flows of this system near the origin, shown in Figure 2.

Analyzing Figure 2, it is clear that only trajectories lying on one curve tend towards the origin. All other trajectories tend arbitrarily far away. Thus, by Theorem 2, the system 22 must undergo a jump transition at $\lambda = \lambda_0$.

CASE 2: Let us look at the case where $b_{11} = 0$ but $b_{12} \neq 0$. This system takes the following form at $\lambda = \lambda_0$:

$$\begin{aligned} y_1' &= ay_2 + \tilde{a}_{11}y_1^2 + \tilde{a}_{12}y_1y_2 + \tilde{a}_{22}y_2^2 + O(y^3) \\ y_2' &= \tilde{b}_{12}y_1y_2 + \tilde{b}_{22}y_2^2 + O(y^3) \end{aligned}$$

We can start by making a simple coordinate transformation. Let $z_1 = y_1 + (\tilde{b}_{22}/\tilde{b}_{12})y_2$ and $z_2 = ay_2$. Now,

$$\begin{aligned} z_1' &= z_2 + \tilde{a}_{11}z_1^2 + a_{12}'z_1z_2 + a_{22}'z_2^2 + O(z^3) \\ z_2' &= \tilde{b}_{12}z_1z_2 + O(z^3) \end{aligned} \tag{35}$$

Here, a_{12}' and a_{22} can be determined from the original coefficients, but this step is not necessary.

We will now use the blow-up method to investigate the degenerate critical point at the origin. First, let us convert our equation to polar coordinates, with $z_1 = r\cos\theta$ and $z_2 = r\sin\theta$. This means that we can use the formulas

$$\begin{aligned} r' &= z_1'\cos\theta + z_2'\sin\theta \\ \theta' &= \frac{z_2'\cos\theta - z_1'\sin\theta}{r} \end{aligned} \tag{36}$$

Making the proper substitutions gives us the following system:

$$\begin{aligned} r' &= r\cos\theta(\sin\theta + \tilde{a}_{11}r\cos^2\theta) + r^2\sin\theta(a_{12}'\cos^2\theta + (a_{22} + \tilde{b}_{12})'\cos\theta\sin\theta) + O(r^3) \\ \theta' &= (\tilde{b}_{12} - \tilde{a}_{11})r\cos^2\sin\theta - \sin^2\theta - r\sin^2\theta(a_{12}'\cos\theta - a_{22}'\sin\theta) + O(r^2) \end{aligned} \tag{37}$$

We are zooming in on the origin, so we want to look at critical points of this new ODE (37) with $r = 0$. Such points must necessarily satsify $\sin\theta = 0$, so the two relevant critical points are $(r, \theta) = (0, 0)$ and $(0, \pi)$. WLOG, let us analyze the critical point at $(0, 0)$.

We can first create a Taylor series approximation of the ODE at $(0, 0)$. Note that the r^2 terms in the expression for θ' all are necessarily multiplied by a factor of $\sin\theta$. This follows from Equation 36 combined with the fact that our original differential equation's 3-jet has no terms in the form $\begin{pmatrix} 0 \\ y_1^3 \end{pmatrix}$ (this term can only exist if $b_{11} \neq 0$). Hence, our transformed differential equation has no terms in the form $\begin{pmatrix} 0 \\ z_1^3 \end{pmatrix}$. Thus, when we expand about $(0, 0)$ (or $(0, \pi)$), the error term in the expression for θ' is $O(|r, \theta|^3)$.

$$r' = r\theta + \tilde{a}_{11}r^2 + O(|r, \theta|^3)$$
$$\theta' = (\tilde{b}_{12} - \tilde{a}_{11})r\theta - \theta^2 + O(|r, \theta|^3)$$

It is clear from this ODE that the critical point at $(r, \theta) = (0, 0)$ is not hyperbolic; thus, we must perform another blow-up. Let us take $r = \rho\cos\phi$ and $\theta = \rho\sin\phi$. Substituting these variables yields:

$$\rho' = \rho^2(\tilde{a}_{11}\cos^3\phi - \sin^3\phi + \cos^2\phi\sin\phi + (\tilde{b}_{12} - \tilde{a}_{11})\cos\phi\sin^2\phi) + O(\rho^3)$$
$$\phi' = \rho\cos\phi\sin\phi((\tilde{b}_{12} - 2\tilde{a}_{11})\cos\phi - 2\sin\phi) + O(\rho^2)$$

We can also divide the field by ρ: this will not change the appearance of the phase portrait, except at $\rho = 0$. After doing this, we get the following system:

$$\begin{aligned} \rho' &= \rho(\tilde{a}_{11}\cos^3\phi - \sin^3\phi + \cos^2\phi\sin\phi + (\tilde{b}_{12} - \tilde{a}_{11})\cos\phi\sin^2\phi) + O(\rho^2) \\ \phi' &= \cos\phi\sin\phi((\tilde{b}_{12} - 2\tilde{a}_{11})\cos\phi - 2\sin\phi) + O(\rho) \end{aligned} \tag{38}$$

Once again, we want to find critical points of this system (38) with $\rho = 0$. Analyzing the system shows that as long as $2\tilde{a}_{11} - \tilde{b}_{12} \neq 0$, there are six hyperbolic fixed points, corresponding to $\phi = 0, \pi/2, \pi, 3\pi/2, \arctan((\tilde{b}_{12} - 2a_{11})/2)$, and $\pi + \arctan((\tilde{b} - 2\tilde{a})/2)$. Since these fixed points are hyperbolic, the Hartman-Grobman theorem states that the behavior of the phase portrait near each equilibrium point is topologically independent of higher order terms $O(|\rho, \phi|^2)$. A detailed diagram of this blow-up method is shown in Figure 3.

It is important to note that the $O(\rho)$ term does not alter the hyperbolicity of the fixed points, but it can result in a shearing of the phase plane near each equilibrium point. In the final blow-up of Figure 3, this would cause the straight lines in the figure to not point directly inwards and outwards. However, this effect will not qualitatively change the phase portrait after blowing down. Thus, we can ignore this $O(\rho)$ term in our blow-up analysis.

In the blow-up method, the topological structure of flows near the origin can be completely determined by blowing-down [Tak74], which is equivalent to reversing each step of the blow-up process. Since the behavior of the flows of (38) near the origin is independent of higher-order terms, then so is the behavior of the initial (coordinate-transformed) system (35).There are six unique flow types based on the quadratic coefficients, which are displayed in Figure 4.

By the same reasoning that was presented in case 1, when $a_{11}b_{12} < 0$, the dynamic transition must be jump at $\lambda = \lambda_0$. This can be deduced by looking at the middle panels in

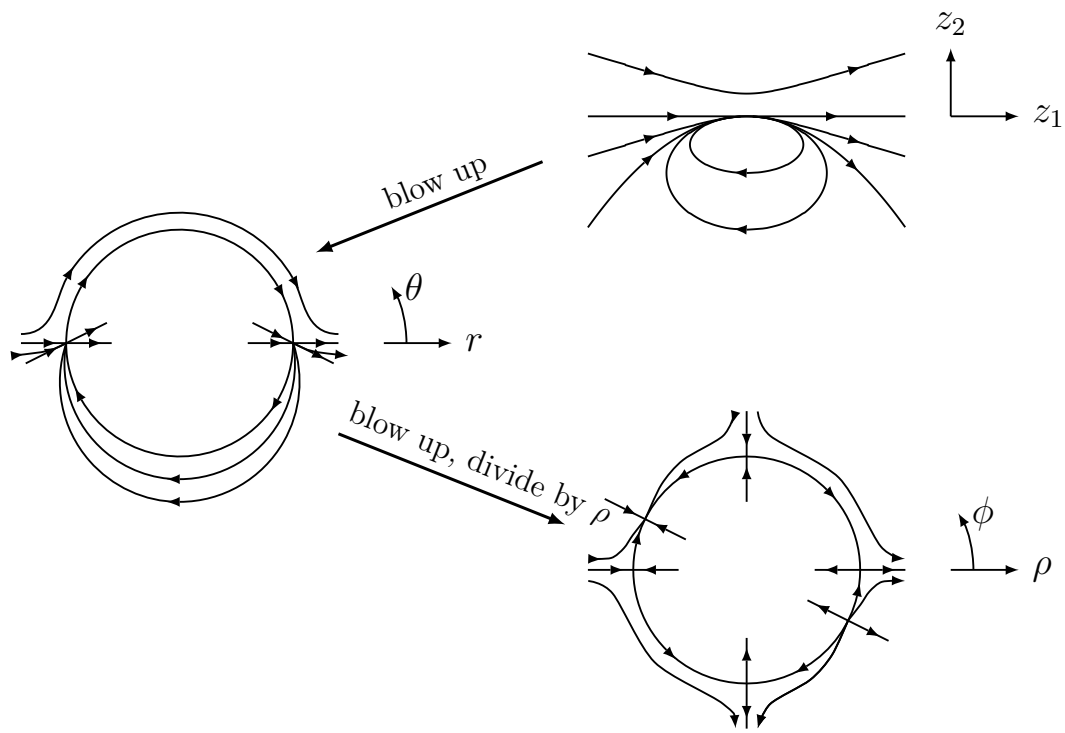

FIGURE 3. Illustration of the blow-up method for case 2 of Theorem 5. The large circles in each of the two blown-up phase portraits represent $r = 0$ and $\rho = 0$, respectively.

Figure 4, which represent the behavior of the system at $\lambda = \lambda_0$. Since the set of trajectories that tend to the origin has an empty interior, the transition at $\lambda = \lambda_0$ is jump by Theorem 2. When $a_{11}b_{12} > 0$, Figure 4 shows that there exists a stable and an unstable region near the origin at $\lambda = \lambda_0$; thus, Theorem 2 allows us to rule out the possibility of a continuous transformation, meaning that the system undergoes either a mixed or jump transition at $\lambda = \lambda_0$.

□

In the previous proof, we saw that there was always a stable region and an unstable region at $\lambda = \lambda_0$ when $a_{11}b_{12} > 0$. This behavior of the reduced ODE at $\lambda = \lambda_0$ suggests the existence of a mixed transition, where trajectories in that stable region converge to a local attractor near the origin when $\lambda > \lambda_0$. An illustration of what this transition would look like is shown in Figure 5. More is needed to prove this fact; in fact, we need to prove the converse of assertion 3 of Theorem 2 in the specific case of Theorem 5, case 2. We can compile these possibilities into a conjecture:

Conjecture 3. *For the parabolic PDE system (22), in the case where $b_{11} = 0$, $a_{11} \neq 0$, $b_{12} \neq 0$, $2a_{11} - b_{12} \neq 0$, $a_{11}b_{12} > 0$, the system undergoes a mixed transition.*

For many combinations of parameters, such as when $b_{11} = b_{12} = 0$, the behavior of the phase portrait for the reduced system depends on fourth order and higher terms - terms that cannot be determined by our leading-order approximation of the center manifold. The insufficiency of a leading-order center manifold approximation suggests a need to find a higher-order approximation of the center manifold in order to generate a more comprehensive

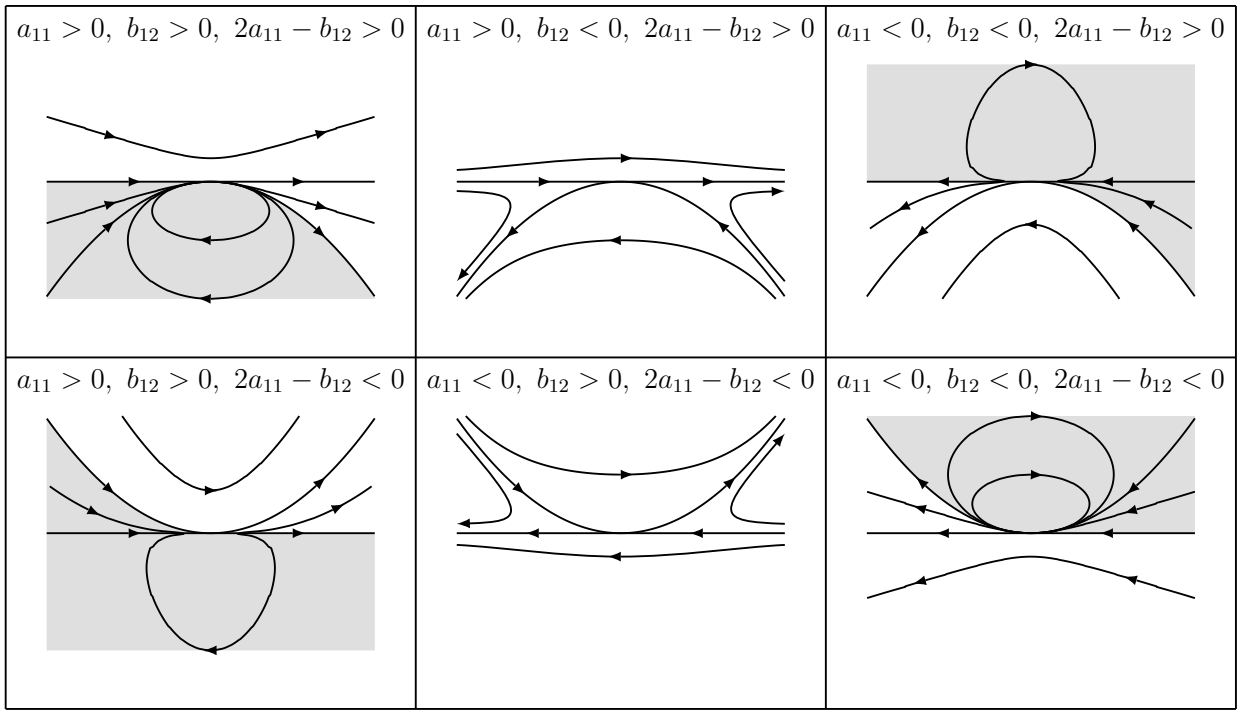

FIGURE 4. Phase portraits of the reduced equation at $\lambda = \lambda_0$ for all possible sub-cases in case 2 of Theorem 5. The shaded regions represent those where trajectories tend to the origin.

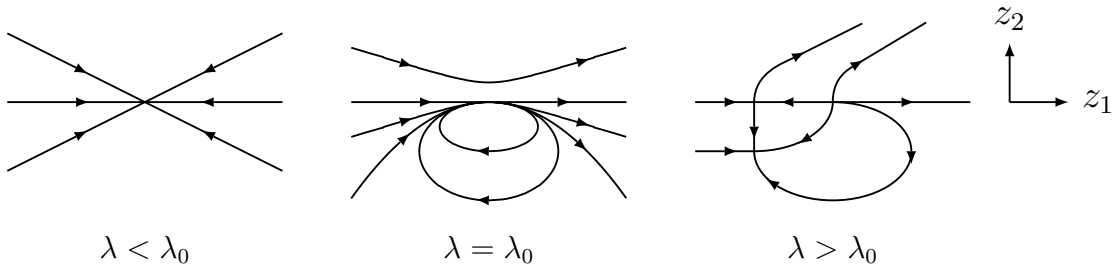

FIGURE 5. Phase portraits for a proposed mixed transition of the parabolic PDE system for $b_{11} = 0$, $a_{11} > 0$, $b_{12} > 0$, and $2a_{11} - b_{12} > 0$.

classification of transition types in the PDE system (22). Applying computational methods such as the backwards-forwards approach could yield a better approximation of the center manifold to resolve this problem.

APPENDIX

Reduced Equation of the Parabolic PDE System. Earlier, we found an explicit reduced equation for our PDE System (22) at the critical threshold $\lambda = \lambda_0$. To make this reduced ODE valid for other values of λ, we must add λ-dependence in the third order terms. When this is done, we get a system that looks very similar to the one presented earlier; the quadratic terms are exactly the same, and the cubic terms are the same, but the

coefficients C_i now depend on λ.

$$
\begin{aligned}
y_1' &= (\lambda-1)y_1 + ay_2 + \frac{8}{3\pi}(a_{11}y_1^2 + a_{12}y_1y_2 + a_{22}y_2^2) + \Big(C_0(\lambda)(2a_{11}^2 + a_{12}b_{11}) \\
&+ 2C_1(\lambda)aa_{11}b_{11} - \frac{3}{4}K_1\Big)y_1^3 + \Big(C_0(\lambda)(3a_{11}a_{12} + a_{12}b_{12} + 2a_{22}b_{11}) \\
&+ C_1(\lambda)a(-4a_{11}^2 + 2a_{11}b_{12} - a_{12}b_{11}) - 4C_2(\lambda)a^2a_{11}b_{11}\Big)y_1^2y_2 \\
&+ \Big(C_0(\lambda)(2a_{11}a_{22} + a_{12}^2 + a_{12}b_{22} + 2a_{22}b_{12}) + C_1(\lambda)a(-4a_{11}a_{12} + 2a_{11}b_{22} - 4a_{22}b_{11}) \\
&+ C_2(\lambda)a^2(2a_{11}^2 - 2a_{11}b_{12} - a_{12}b_{11}) + 2C_3(\lambda)a^3a_{11}b_{11}\Big)y_1y_2^2 + \Big(C_0(\lambda)(a_{12}a_{22} + 2a_{22}b_{22}) \\
&+ C_1(\lambda)a(-a_{12}^2 + a_{12}b_{22} - 2a_{22}b_{12}) + C_2(\lambda)a^2(a_{11}a_{12} - a_{12}b_{12} + 2a_{22}b_{11}) + C_3(\lambda)a^3a_{12}b_{11}\Big)y_2^3 \\
&+ o(\|y\|^3) \\
y_2' &= (\lambda-1)y_2 + \frac{8}{3\pi}(b_{11}y_1^2 + b_{12}y_1y_2 + b_{22}y_2^2) + \Big(C_0(\lambda)(2a_{11}b_{11} + b_{11}b_{12}) + 2C_1(\lambda)ab_{11}^2\Big)y_1^3 \\
&+ \Big(C_0(\lambda)(a_{11}b_{12} + 2a_{12}b_{11} + 2b_{11}b_{22} + b_{12}^2) + C_1(\lambda)a(-4a_{11}b_{11} + b_{11}b_{12}) - 4C_2(\lambda)a^2b_{11}^2\Big)y_1^2y_2 \\
&+ \Big(C_0(\lambda)(a_{12}b_{12} + 2a_{22}b_{11} + 3b_{12}b_{22}) + C_1(\lambda)a(-2a_{11}b_{12} - 2a_{12}b_{11} - 2b_{11}b_{22}) \\
&+ C_2(\lambda)a^2(2a_{11}b_{11} - 3b_{11}b_{12}) + 2C_3(\lambda)a^3b_{11}^2\Big)y_1y_2^2 + \Big(C_0(\lambda)(a_{22}b_{12} + 2b_{22}^2) \\
&+ C_1(\lambda)a(-a_{12}b_{12} - b_{12}b_{22}) + C_2(\lambda)a^2(a_{11}b_{12} + 2b_{11}b_{22} - b_{12}^2) + C_3(\lambda)a^3b_{11}b_{12} - \frac{3}{4}K_2\Big)y_2^3 \\
&+ o(\|y\|^3)
\end{aligned}
$$

Now, we can define the $C_i(\lambda)$ as follows:

$$C_i(\lambda) = \sum_{k=1}^{\infty} \frac{(-1)^{i+1}64i!}{\pi^2(\lambda-(2k+1)^2)^{i+1}(2k-1)^2(2k+1)^2(2k+3)^2}$$

Whenever λ is near λ_0, these coefficients $C_i(\lambda)$ will be positive. Plugging in $\lambda = \lambda_0 = 1$ yields the same coefficients C_i and the same reduced equation as in the previous section.

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